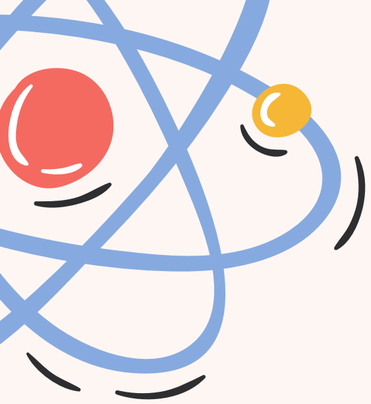
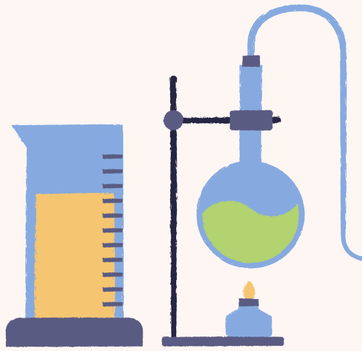
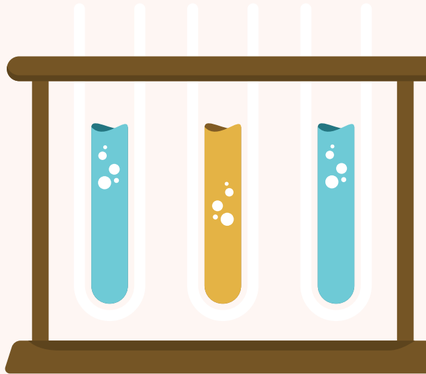
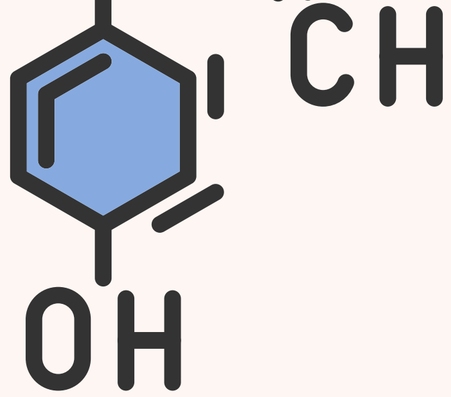




MOLE, MOLAR MASS, EMPIRICAL AND MOLECULAR FORMULA



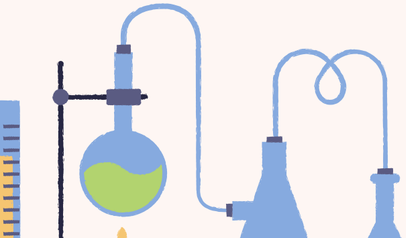


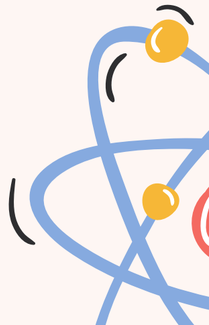




Learning Objectives:

1. define a mole
2. determine the molar mass of elements and compounds; and
3. calculate the mass of a given number of moles or particles of an element or compound, or vice versa.





Mole



1 dozen of eggs

=

how many eggs?

Mole



1 dozen of eggs

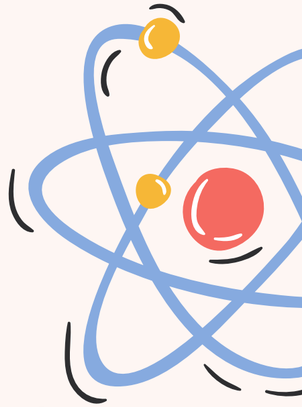
=

12 eggs

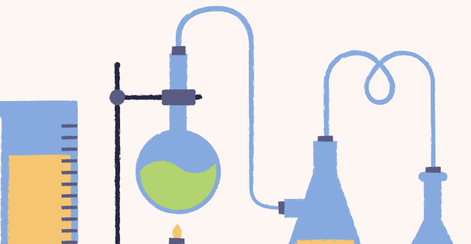
Mole

Is a quantity, it represents a very large number.

Mole



is the “counting unit” used by chemists to indicate the number of atoms, ions, molecules or formula units present in a particular chemical sample.



Amedeo Avogadro



Best known for his work on the discovery of his gas law known as Avogadro's law.

Amedeo Avogadro



This law states that at the same temperature and pressure, equal amounts of different gasses will contain an equal number of particles.

Mole

$$1 \text{ mole} = 6.022 \times 10^{23}$$

(avogadro's number)



1 mole of books = 6.022×10^{23}

Mole

It represents a quantity of atoms or particles.

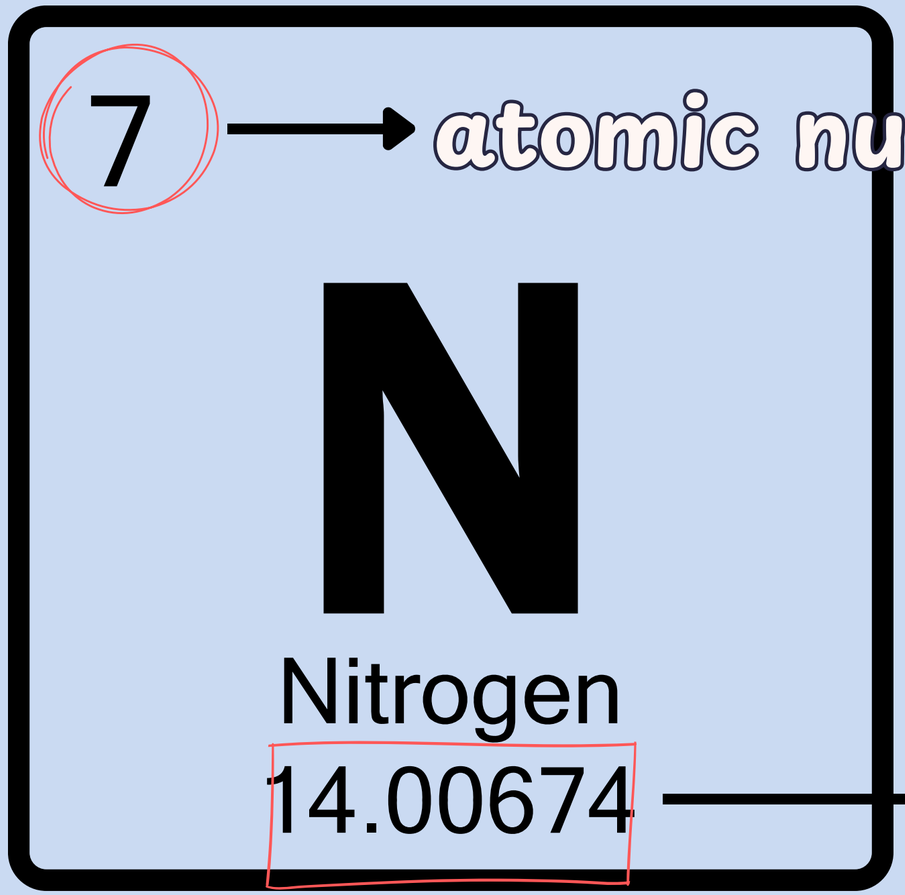
Mole

1 mol C = 6.022×10^{23} atoms of C

2 mol C = $2 \times 6.022 \times 10^{23}$ atoms of C

1 mol CO_2 = 6×10^{23}

Mole



atomic number

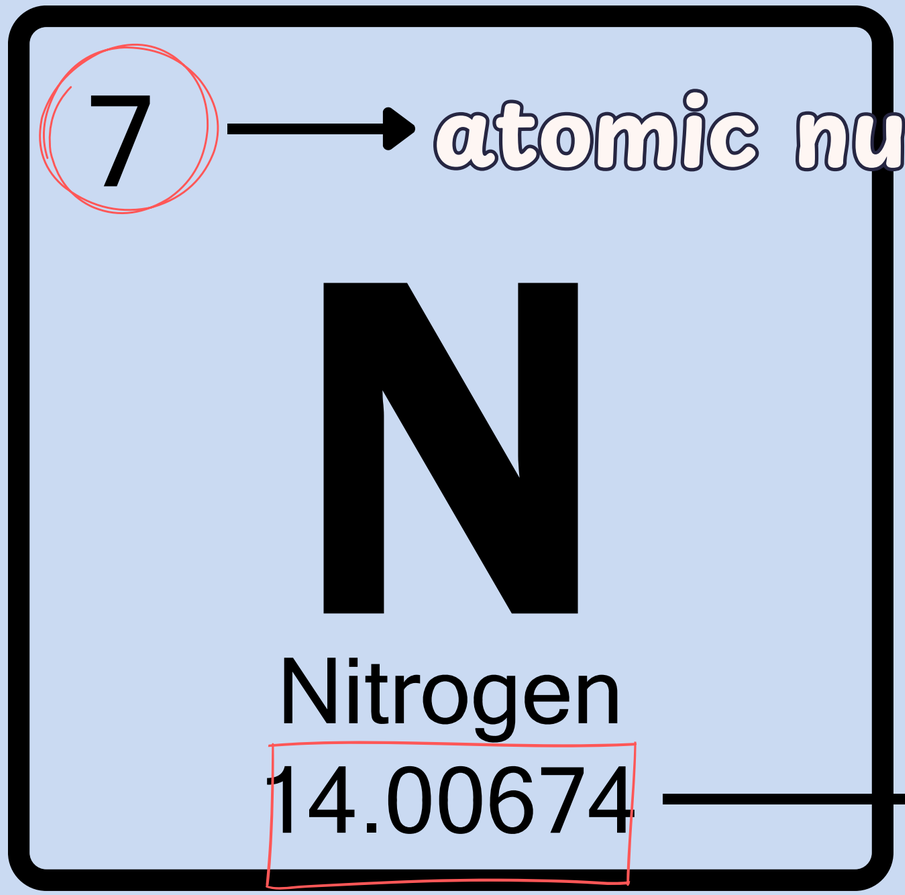
14 atomic mass unit (amu)

14 - represents molar mass

14 g/mol

mass number

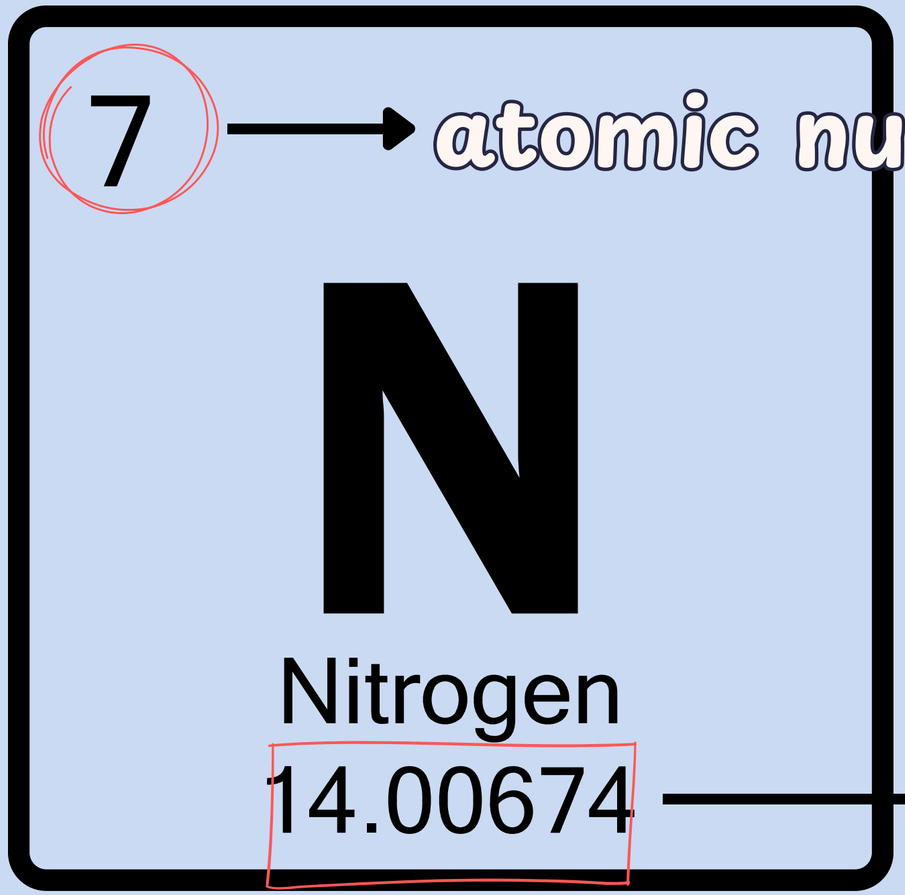
Mole



1 mol N = 14 g N

14 g N = 6×10^{23} atoms

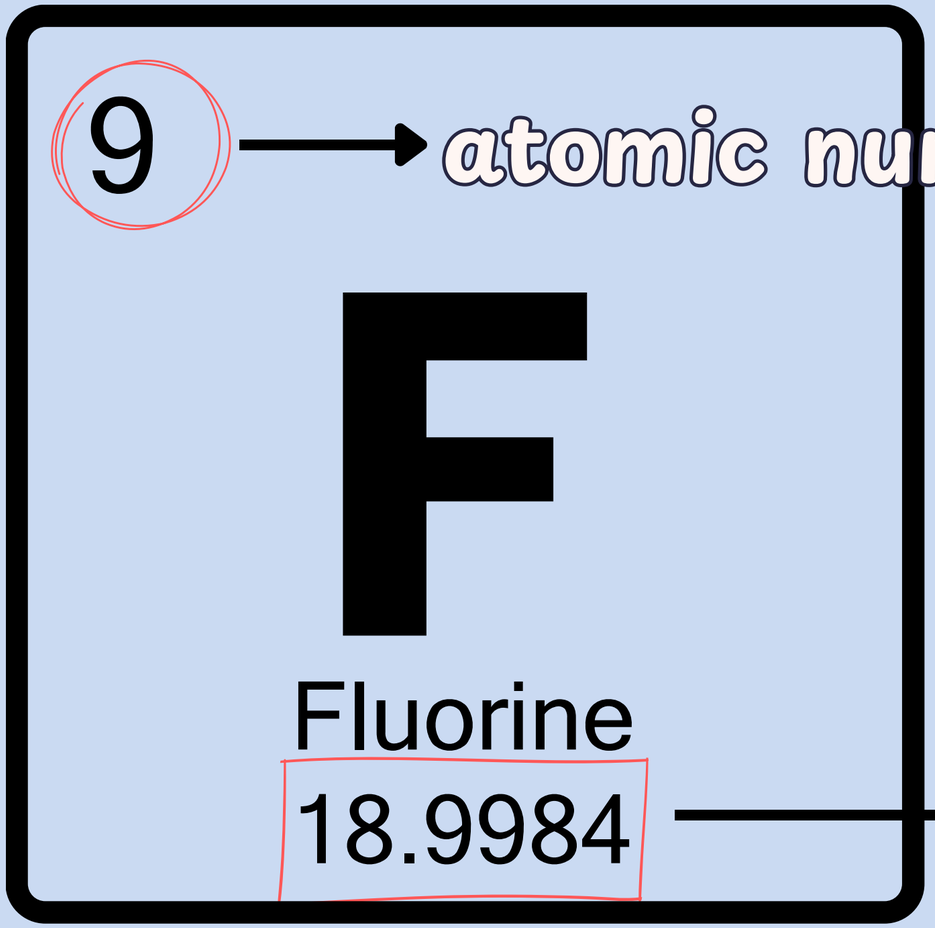
Mole



2 mol N = 14 + 14 = 28 g N

3 mol N = 14 x 3 = 42 g N

Mole

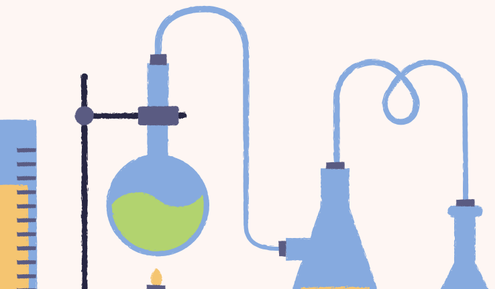


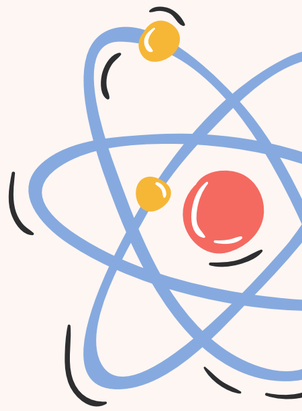
19 g/mol

19 g F = 1 mol = 6×10^{23} atoms

Molar mass

(g/mol) is defined as the mass of a given element or molecule of that substance.





Molar mass is the mass of one mole

6.022×10^{23}

- ✓ particles
- ✓ molecules
- ✓ atoms

Practice:

Get the molar mass of $C_6H_{12}O_6$

Get the molar mass of $Mg(OH)_2$

Get the molar mass of $(NH_4)_3PO_4$

Practice:

Get the molar mass of $C_6H_{12}O_6$

C 6 x 12 = 72

H 12 x 1 = 12

O 6 x 16 = 96

180 g/mol

Practice:

Get the molar mass of $Mg(OH)_2$

Mg: 24.31 x 1 = 24.31

O: 16.00 x 2 = 32

H: 1.01 x 2 = 2.02

58.33 g/mol

Practice:

Get the molar mass of $(NH_4)_3PO_4$

N: 14.01 x 3 = 42.03

H: 1.01 x 12 = 12.12

P: 30.97 x 1 = 30.97

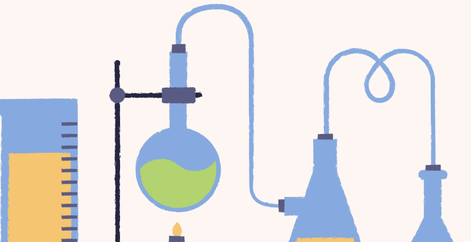
O: 16.00 x 4 = 64

149.12 g/mol

Empirical Formula

(simplest form)

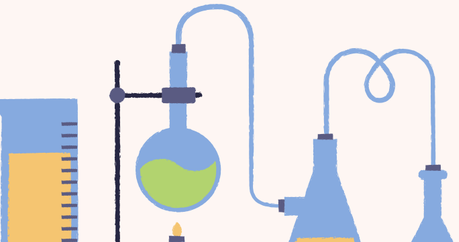
It represents the simplest whole number ratio of various atoms present in a compound.



Molecular Formula

(complex)

It shows the exact number of different types of atoms present in a molecule of a compound.

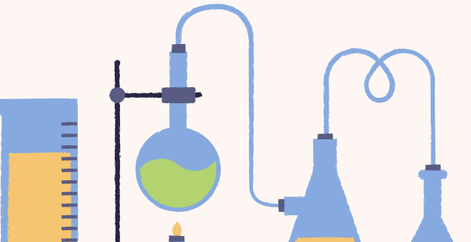


Sample exercise #1

A 0.545 g sample of aluminum burns completely in oxygen, producing 1.030 g of aluminum oxide. Find the empirical formula of the oxide.

(Problem) (Problem) (Periodic Table)

Element	% Composition	Atomic Mass	No. of Moles	Simplest Ratio

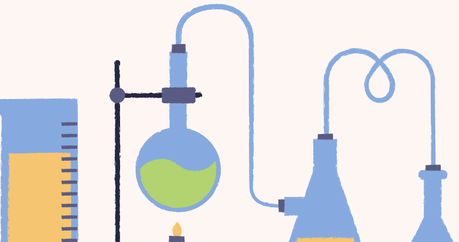


Note:

If you get unusual decimals like 0.5, 0.33, and 0.25, multiply these numbers by the smallest possible number that will convert them into whole numbers.

Sample exercise #2

Calculate the empirical formula of 40.0% C, 6.7% H, 53.3% O



NOTE:

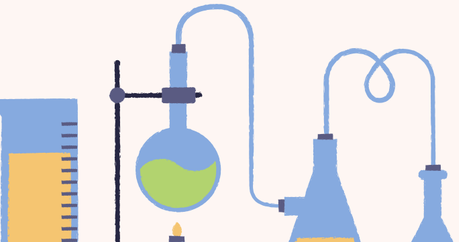
- If the result is very close to a whole number (like 1.996), we round it off instead of multiplying.

So why not multiply 1.996 by 2?

- 1.996 is very close to 2.000 (less than 0.01 away).
- If it were something like 1.5 or 1.33, we would multiply to get rid of the decimal (e.g., ×2, ×3).

Sample exercise #3

Calculate the empirical formula of 70.0% Fe, 30.0% O



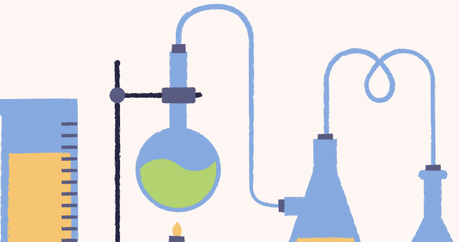
NOTE:

- 1.496 → this is close to 1.5, not a whole number

- Multiply both values by 2 to eliminate the .5

Sample exercise #4

A compound has an empirical formula of CH_2O and a molar mass of 180 g/mol. What is the molecular formula?



Sample exercise #5

Given the percentage composition as C - 24.27, H- 4.07, and Cl -71.65, calculate the empirical formula given the molar mass of the compound as 98.98 g.

(Problem) (Problem) (Periodic Table)

Element	% Composition	Atomic Mass	No. of Moles	Simplest Ratio

